

SETTING UP A VIRTUAL HOME LAB

I built a free virtual IT lab from scratch on my computer using open-source software.

In the early 2000s the only way one could have built an IT lab was to purchase physical servers or re-purpose old computers. While this can be greatly beneficial as far as IT experience and skills, it can be much faster simply to build a free virtualized IT lab.

Let us dive in!

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Virtualization Overview

is virtualization So what anyway? The IT term for virtualization refers to the practice of emulating (or stimulating) a computer. You can use a virtual machine to run a computer as an application on your host computer.

Virtual machines act like real computers but run as a software program. So, you can start, restart, shut down, install software and operating systems just like you would on a normal computer.

This makes virtual machines perfect for IT labs.

Free Virtualization Software

With so many different virtualization software options available, it can be confusing to know which one is the “best” and which one you should use. Let us address this now.

There is no “best” virtualization software. There are simply different options and the ones you should consider will depend on your host operating system.

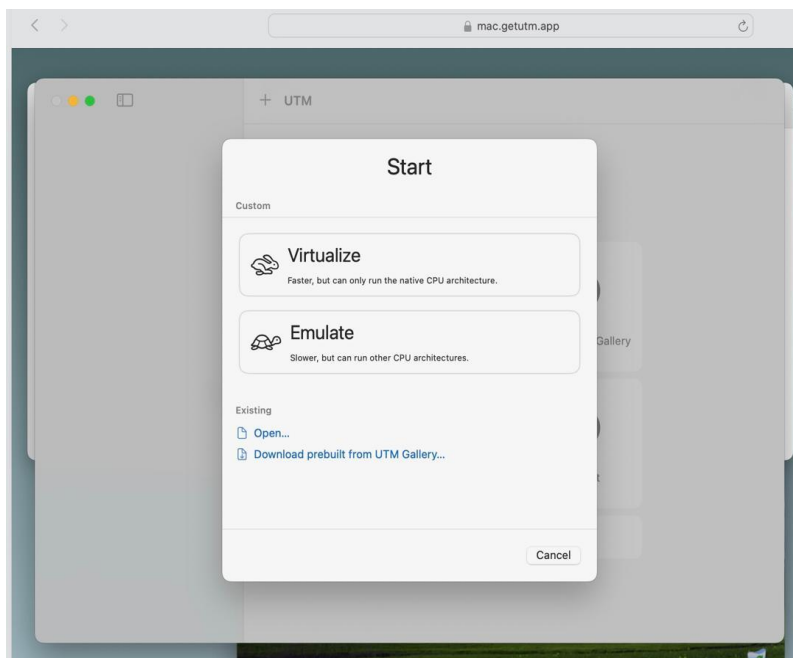
In this setup I will be using UTM because it is compatible with both window and Linux, but you can use anyone that suit your computer. This step will be identical to other virtualization software.

Downloading and installing a UTM

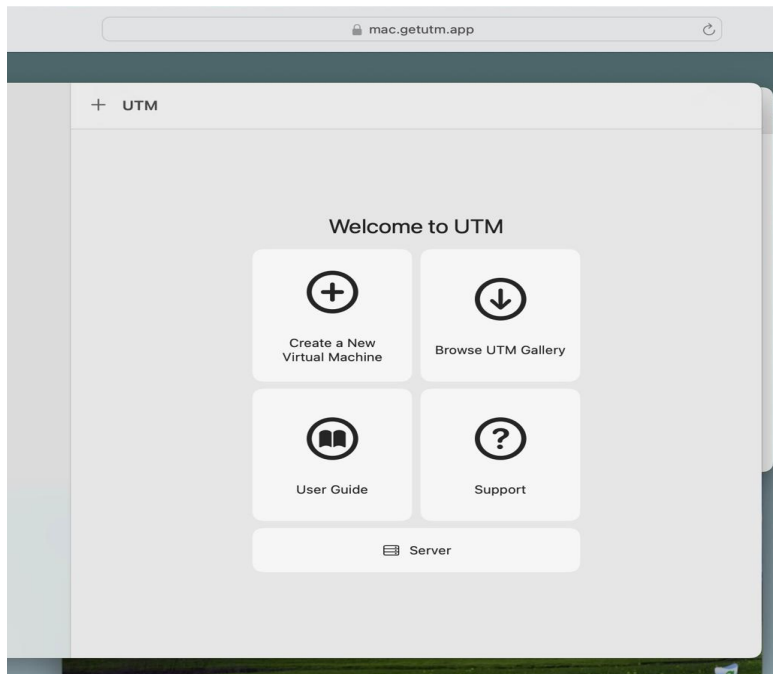
Download UTM by visiting the website <http://mac.getutm.app>. Simply click on the “Download” button to initiate the download process, then proceed to install it on your Mac once download is completed.



I proceeded through installation by clicking on the option virtualize.



Click virtualize and launch your UTM.



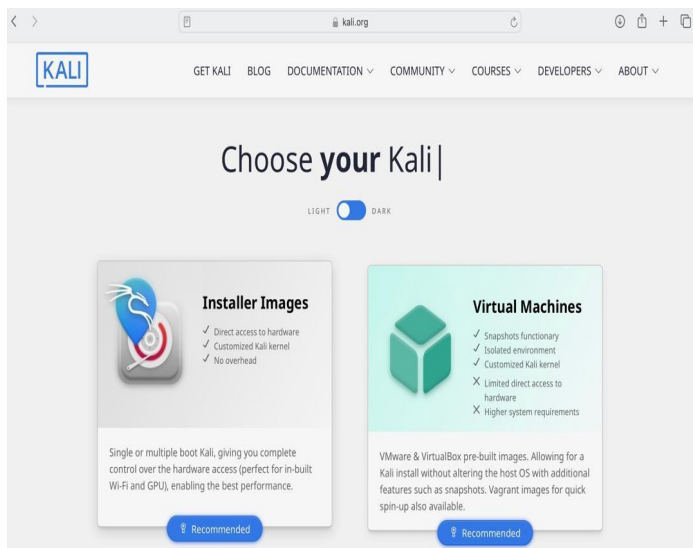
Creating a virtual Machine

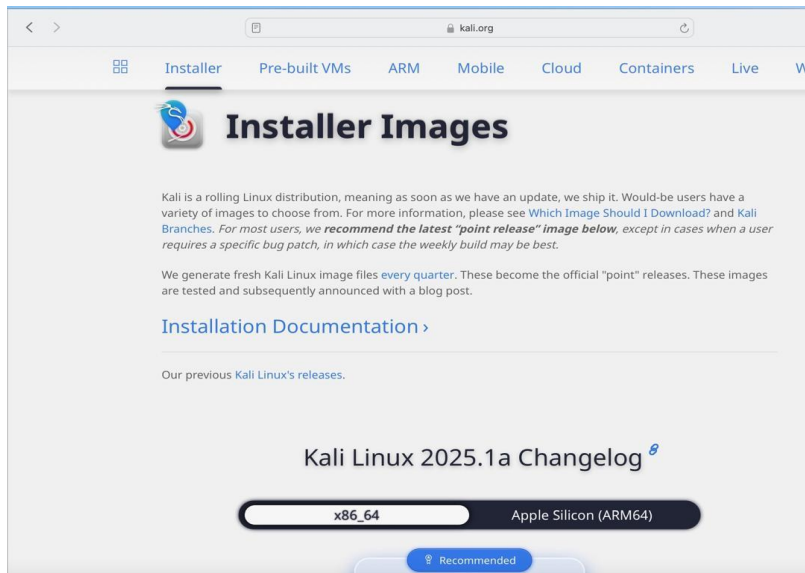
Kali Linux

Kali Linux is a specialized Linux distribution designed mainly for cybersecurity tasks like penetration testing, ethical hacking and security auditing.

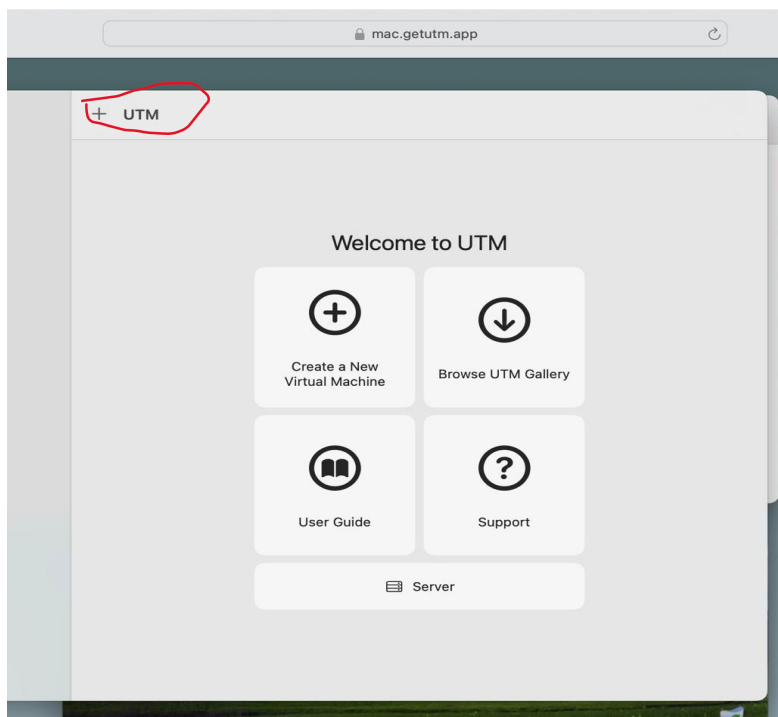
Now that we have our UTM, we can then go ahead and download our Kali Linux. To download Kali Linux, visit the website: <https://www.kali.org/get-kali/>.

- choose the installer image.
- X86_64
- Download

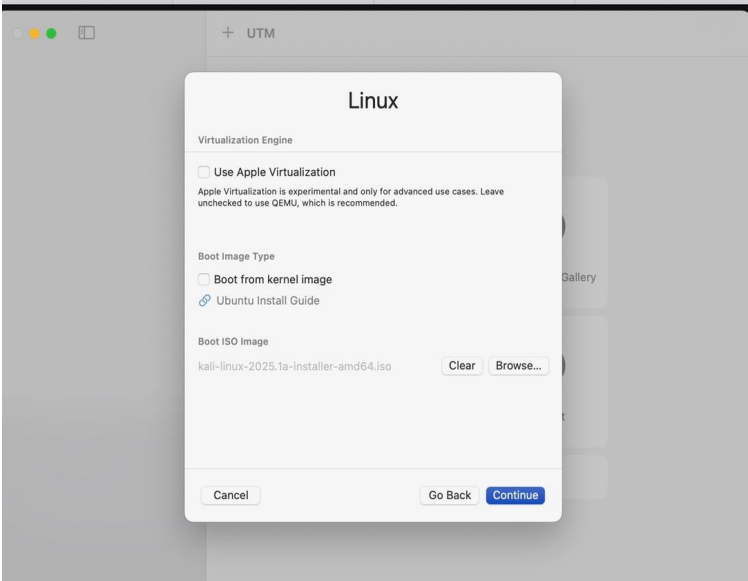
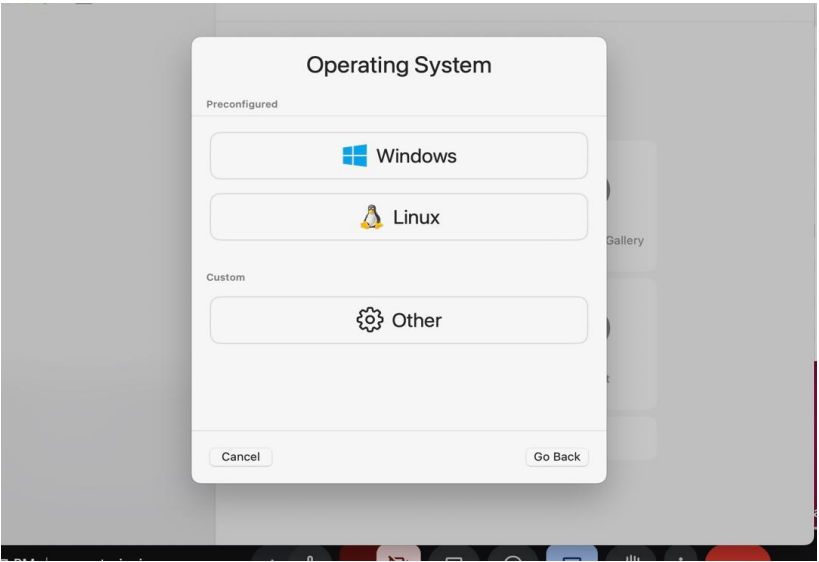
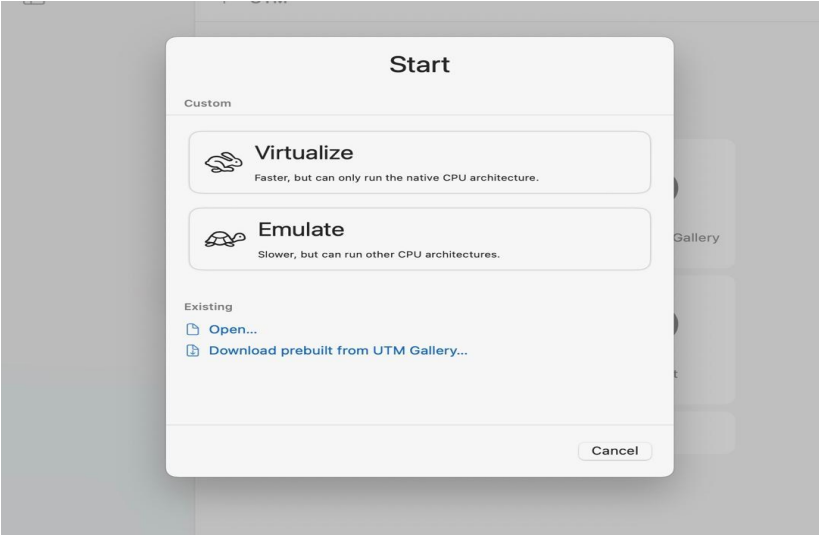




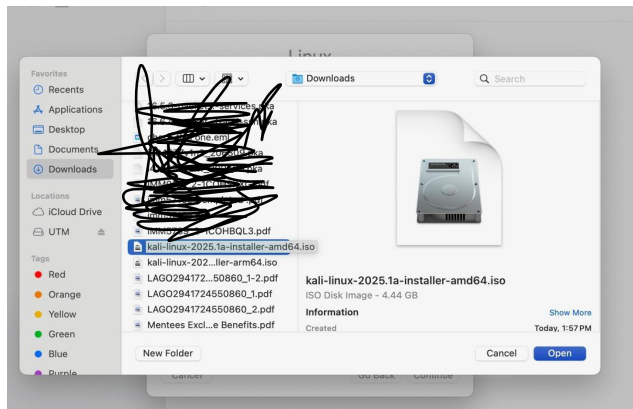
After downloading, go to your UTM to mount Kali Linux by clicking on the + sign.



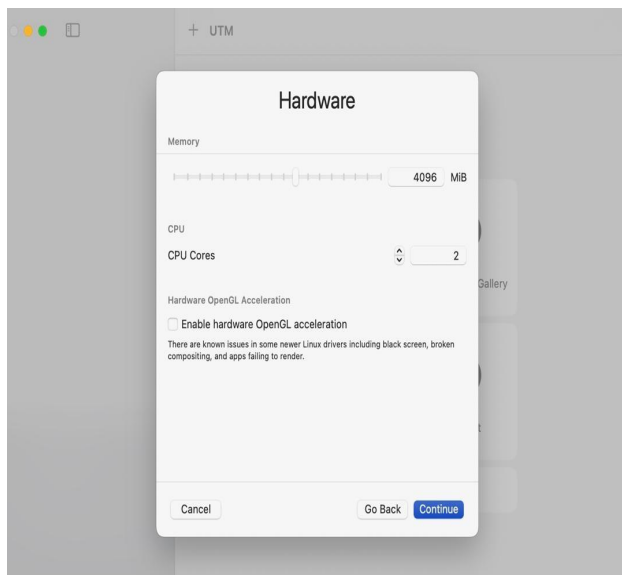
1. click virtualize.
2. Click on Linux
3. Click on browse.



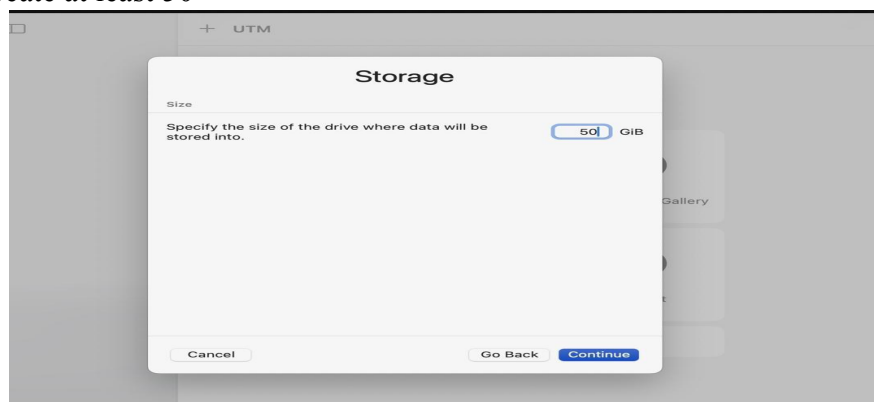
It takes you to your downloads, double click on Kali Linux



- Click open it takes back to where you clicked on browse and then click on continue.
- Allocate at least 4000 RAM.
- And 2 CPU core

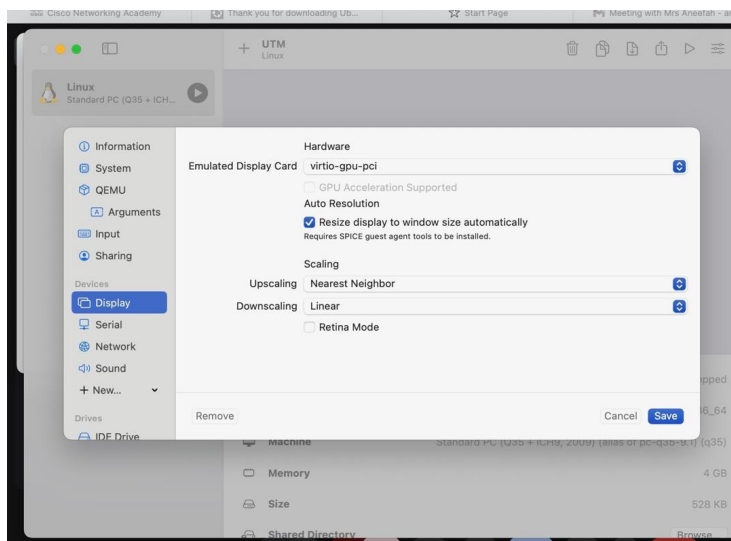
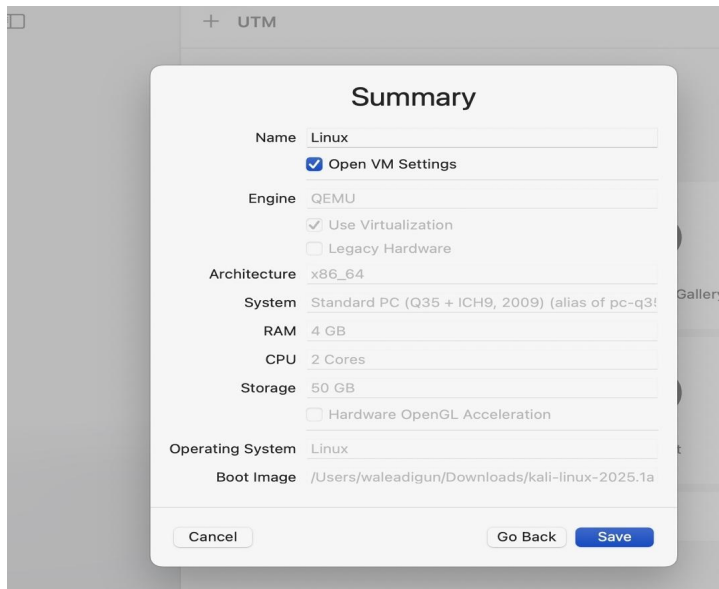


- Click on continue.
- Allocate at least 50

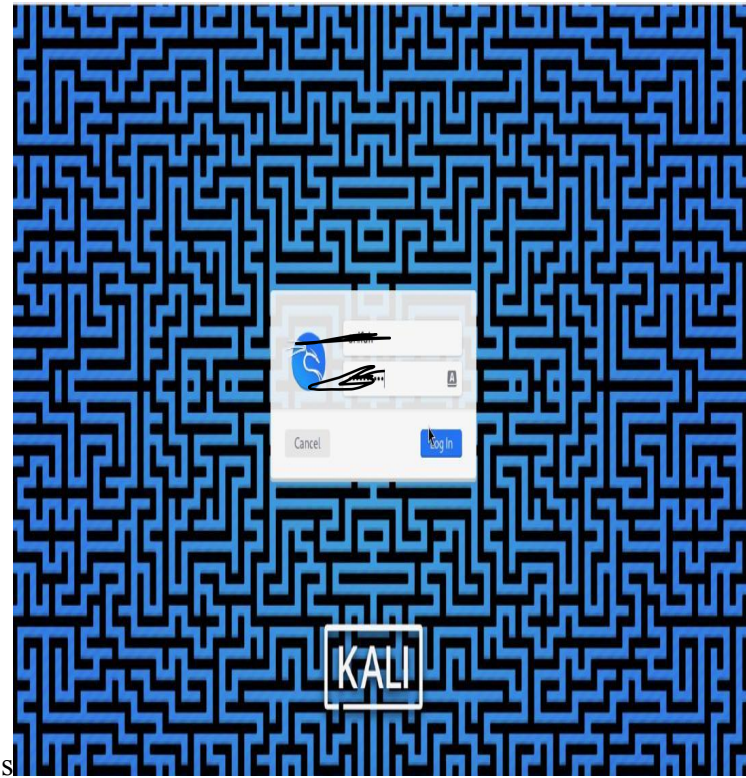
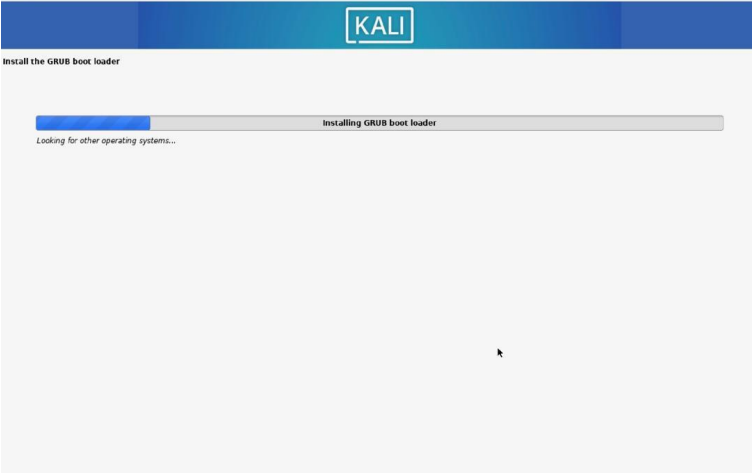


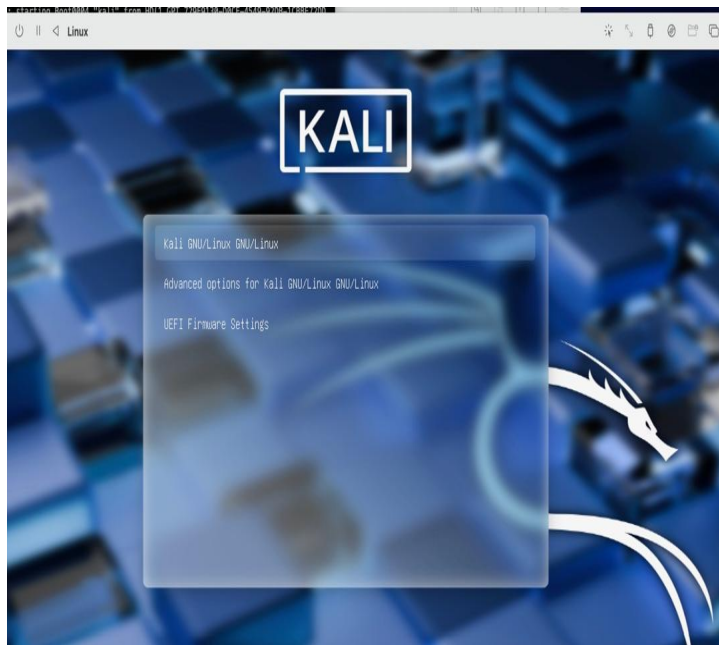
GiB.

- Click on continue.
- Click on save.



- Set up language.
- Configure network (Domain name)
- Partition disk (use entire disk)
- Partition disk (all files in one partition)
- Finish installation

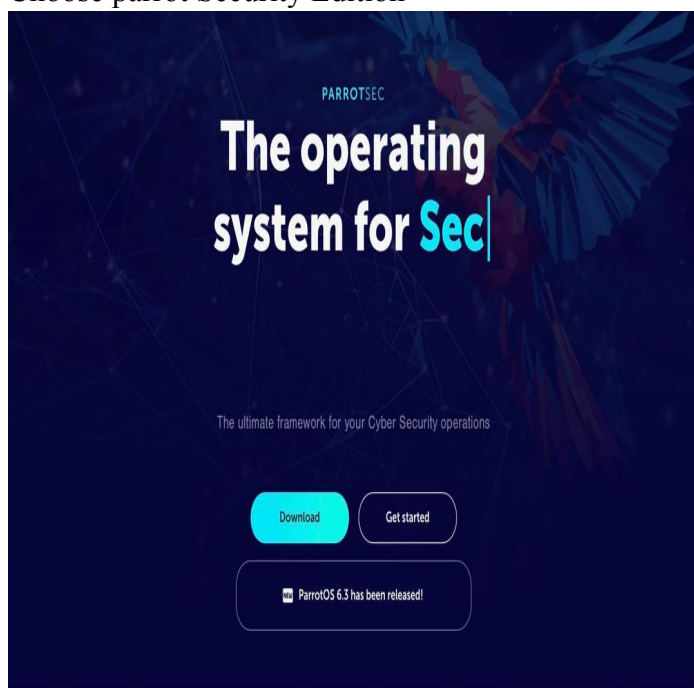


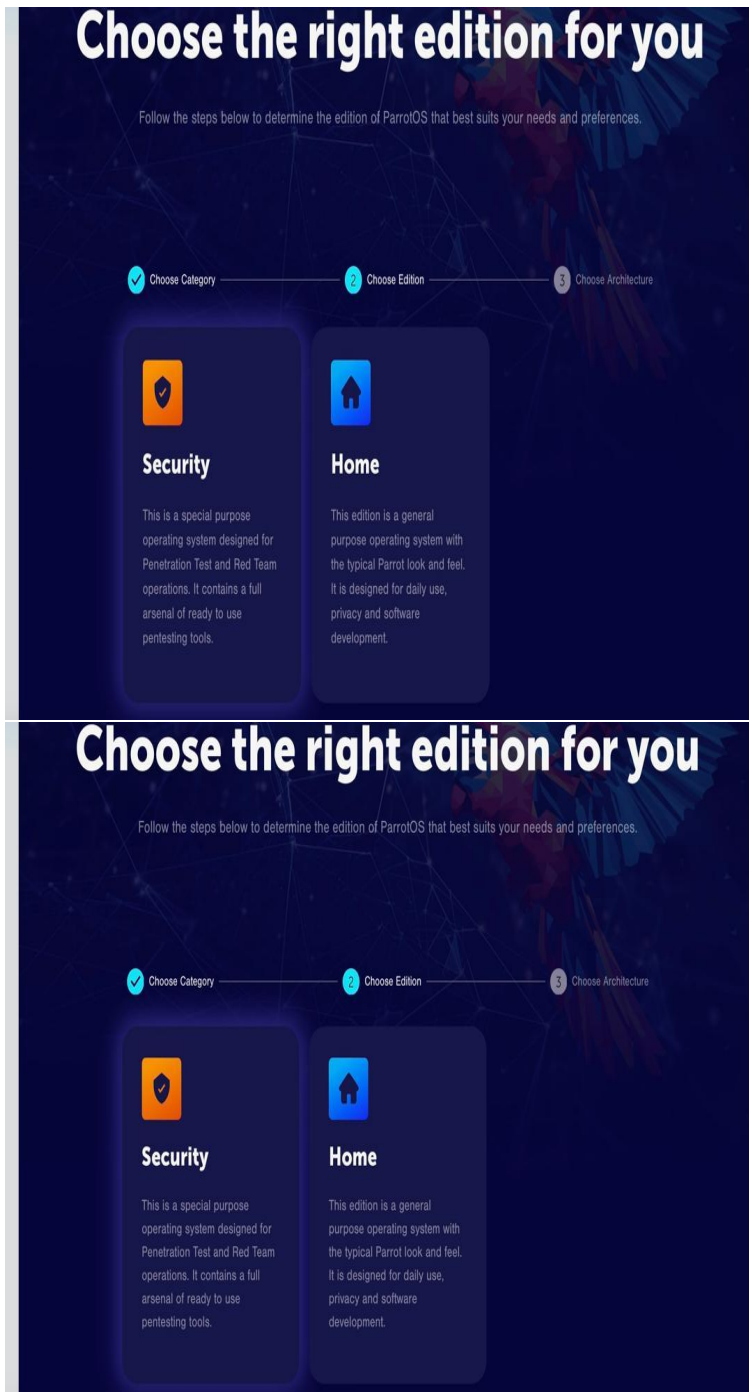


Parrot Security

This is another great ethical hacking tool like Kali Linux but it's lightweight and privacy focused.

1. Visit the link <http://www.parrotsec.org/download/>
2. Choose parrot Security Edition





- Go to your download
- Double click on parrot
- Parrot is installed on UTM

